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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEMBG201 - COMPUTATIONAL BIOLOGY AND BIOINFORMATICS

Part A

Very Short Answer Questions

Answer all questions

Each question carries 1 mark

1. **Explain the purpose of nucleotide sequence databases.**

Nucleotide sequence databases store, organize, and provide access to DNA and RNA sequence data for biological research and analysis.

2. **Define the term "Knowledge Discovery in Databases (KDD)."**

Knowledge Discovery in Databases (KDD) is the process of extracting useful, valid, and previously unknown information or patterns from large datasets.

3. **What was the main goal behind developing Python?**

Python was developed to provide a simple, readable, and easy-to-learn programming language that improves programmer productivity.

4. **What is flexible docking?**

Flexible docking is a molecular docking method that allows the ligand and/or receptor to change conformation during the docking process to achieve better binding predictions.

5. **Name any one ligand database.**

PubChem.

Part B

Short Answer Questions

Answer any 10 questions

Each question carries 2 marks

6. **Describe agglomerative clustering.**

Agglomerative clustering is a hierarchical clustering method in which each data point starts as a separate cluster, and clusters are progressively merged based on similarity until a single cluster or the desired number of clusters is obtained.

7. What is meant by portability in Python?

Portability in Python refers to the ability of Python programs to run on different operating systems with little or no modification.

8. Explain the concept of Information Gain in the context of tree induction.

Information Gain is a measure used in decision tree induction to determine the best attribute for splitting data. It represents the reduction in entropy (uncertainty) achieved by partitioning the dataset based on an attribute.

9. Distinguish between native and denatured proteins.

- Native protein: A protein in its natural, biologically active three-dimensional structure.
- Denatured protein: A protein whose structure has been altered by heat, chemicals, or pH changes, resulting in loss of biological activity.

10. State two applications of XML.

1. Data storage and exchange between different systems.
2. Development of web services and configuration files.

11. State any two features of Python strings.

1. Strings are immutable (cannot be changed after creation).
2. Strings support indexing and slicing operations.

12. What is virtual screening?

Virtual screening is a computational technique used to evaluate large libraries of compounds and identify potential drug candidates that are likely to bind to a target protein.

13. What is a biological database?

A biological database is an organized collection of biological information, such as DNA sequences, protein structures, genes, and biochemical data, stored electronically for easy access and analysis.

14. What is meant by database integration in PHP?

Database integration in PHP refers to connecting PHP applications with databases (such as MySQL) to store, retrieve, update, and manage data dynamically.

15. What is the importance of indentation in Python files?

Indentation in Python defines code blocks and program structure. Proper indentation is mandatory for correct execution and improves code readability.

16. Define binding site prediction.

Binding site prediction is the computational process of identifying regions on a biomolecule, usually a protein, where ligands or other molecules can bind.

17. What is meant by gene annotation?

Gene annotation is the process of identifying genes within a DNA sequence and assigning information about their structure, function, and biological significance.

Part C

Short Essay Questions

Answer any 5 questions

Each question carries 6 marks

18. Describe how secondary databases help researchers.

Any 6 functions of Secondary Databases – 6 marks

19. Explain how primary sequence of amino acids contributes to the 3D function of a protein.

Role in Protein Structure and Function – 6 marks

1. Determines Protein Folding

The amino acid sequence directs the folding of the protein into a specific 3D structure.

2. Formation of Secondary Structures

Hydrogen bonding between amino acids forms α-helices and β-sheets.

3. Establishes Tertiary Structure

Interactions among side chains produce the final three-dimensional conformation.

4. Creates Active Sites

Specific amino acid arrangements form catalytic and binding sites.

5. Influences Stability

Hydrophobic interactions, ionic bonds, and disulfide bridges stabilize the structure.

6. Mutations Affect Function

Changes in amino acid sequence can alter folding and biological activity.

20. Explain the role of web log files in web usage mining.

Role of Web Log Files – 6 marks

1. User Behavior Analysis

Track pages visited, time spent, and navigation patterns.

2. Session Identification

Help identify individual user sessions.

3. Traffic Analysis

Provide information about website popularity and visitor trends.

4. Personalization

Support recommendation systems and customized content delivery.

5. Website Optimization
Identify frequently visited pages and improve website design.
6. Security Monitoring
Detect suspicious activities and unauthorized access attempts.

21. Explain the role of KEGG in pathway analysis and functional genomics.
Any 6 functions of KEGG – 6 marks
1. Pathway Mapping
Provides graphical representations of metabolic and signaling pathways.
2. Gene Function Annotation
Links genes to biological pathways and cellular processes.
3. Understanding Biological Systems
Helps study interactions among genes, proteins, and metabolites.
4. Disease Analysis
Identifies pathways involved in diseases and disorders.
5. Drug Discovery
Supports identification of therapeutic targets and drug mechanisms.
6. Comparative Genomics
Allows comparison of pathways across different organisms.

22. Describe hierarchical clustering and its application in gene expression analysis.
Definition – 1 mark, working principle-2 marks, applications – 3 marks
Hierarchical clustering is an unsupervised learning method that groups similar data into clusters.
Working Principle
1. Similarity Calculation
Measures similarity between genes or samples using distance metrics.
2. Cluster Formation
Similar genes are grouped together step by step.
3. Dendrogram Construction
Results are displayed as a tree-like diagram called a dendrogram.
4. Agglomerative Approach
Starts with individual genes and merges similar clusters.
5. Divisive Approach
Starts with one cluster and divides it into smaller clusters.
Applications in Gene Expression Analysis
6. Identification of Co-expressed Genes
Genes with similar expression patterns are grouped together.
7. Sample Classification
Helps classify tissues, diseases, and treatment responses.
8. Functional Prediction

Genes within the same cluster often perform related functions.

23. Describe the key assumptions behind the Naive Bayes classifier.

Naive Bayes is a probabilistic classification algorithm based on Bayes' theorem.

Key Assumptions – 6 marks

1. Conditional Independence

Each feature contributes independently to the outcome.

2. Equal Importance of Features

All predictor variables are considered equally important.

3. Presence of Class Variable

The dataset contains predefined class labels.

4. No Feature Interaction

Relationships between features are ignored.

5. Probability-Based Classification

Classification is based on posterior probabilities.

6. Feature Values are Independent Within Classes

Feature values are assumed independent when the class is known.

Advantages

Simple, fast, and effective for large datasets and text classification.

24. Explain structure-based and ligand-based drug design approaches with suitable examples.

Structure-Based Drug Design (SBDD) – 3 marks

Ligand-Based Drug Design (LBDD) – 3 marks

Part D

Essay Questions

Answer any 1 question

Each question carries 15 marks

25. Explain the Quaternary Structure of Proteins with Suitable Examples (15 Marks)

Definition – 2 marks

Characteristics of Quaternary Structure – 4 marks

Stabilizing Interactions – 3 marks

Formation of Quaternary Structure – 2 marks

1. Individual polypeptide chains fold into tertiary structures.
2. Folded subunits interact through non-covalent forces.
3. The assembled complex forms a stable and functional protein.

Biological Importance – 1 mark

Example – 3 marks

26. Explain the Importance of Minimum Support and Confidence Thresholds in Rule Mining (15 Marks)

Association Rule Mining – 2 marks

Minimum Support- Importance of Minimum Support – 3 marks

Minimum Confidence -Importance of Minimum Confidence – 3 marks

Relationship Between Support and Confidence – 3 marks

Role in Association Rule Mining Algorithms – 2 marks

Applications – 2 marks